COMPUTER ENGINEERING

QUEZON CITY

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| **COURSE CODE/TITLE:** | CPE 010 Data Structures and Algorithms | **SECTION:** | CPE 21S4 |
| **DATE PERFORMED:** | 21/09/2026 | **DATE SUBMITTED:** | 21/09/2026 |
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**ACTIVITY NO. ___** Hands-on Activity 6.1 Searching Techniques

**1. PROCEDURE OUTPUT**

I used VSA code and sorted arrays

**2. ANSWERS TO SUPPLEMENTAL ACTIVITY**

## Answer the following questions:

### 1. What is sorting algorithm?

A sorting algorithm is a way of arranging data in a certain order. For example, numbers can be arranged from smallest to largest or largest to smallest. Names can also be arranged alphabetically. Sorting makes information easier to organize, understand, and find.

### 2.Where can sorting algorithms be used?

Sorting algorithms can be used in many situations. For example, they can be used to arrange student grades from highest to lowest, organize names alphabetically, sort numbers, or arrange items in a list. Sorting is useful because it helps keep data organized and makes information easier to find.

### 3. Explain the different types of sorting algorithms.

There are different types of sorting algorithms, such as Selection Sort, Insertion Sort, and Bubble Sort. Selection Sort finds the smallest number and puts it in the correct position. Insertion Sort takes one number at a time and places it where it belongs in the sorted part of the list. Bubble Sort compares two numbers next to each other and switches them if they are in the wrong order. They all have the same goal, which is to arrange data, but they do it in different ways.

## 4. Give sample programs in C++ that uses sorting algorithms, specifically selection sort, insertion sort, and bubble sort. Explain how the programs work.

**Selection sort:** Finds the smallest element in the unsorted part and places it at the correct position. Repeats until the list is sorted.

```cpp
#include <iostream>
using namespace std;

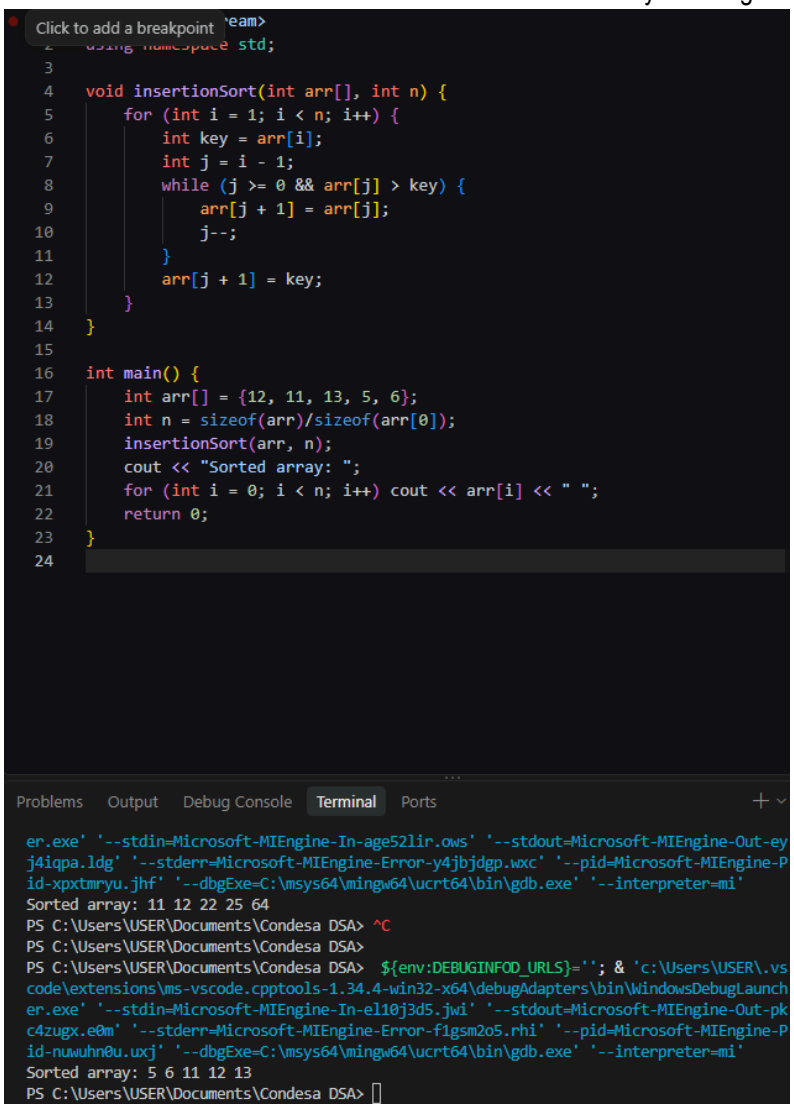
void selectionSort(int arr[], int n) {
    for (int i = 0; i < n - 1; i++) {
        int minIndex = i;
        for (int j = i + 1; j < n; j++) {
            if (arr[j] < arr[minIndex]) {
                minIndex = j;
            }
        }
        swap(arr[i], arr[minIndex]);
    }
}

int main() {
    int arr[] = {64, 25, 12, 22, 11};
    int n = sizeof(arr)/sizeof(arr[0]);
    selectionSort(arr, n);
    cout << "Sorted array: ";
    for (int i = 0; i < n; i++) cout << arr[i] << " ";
    return 0;
}
```

```
Problems  Output  Debug Console  Terminal  Ports

er.exe' '--stdin=Microsoft-MIEngine-In-25eofm42.ild' '--stdout=Microsoft-MIEngine-Out-xi
gvszcv.el1' '--stderr=Microsoft-MIEngine-Error-1te1qovx.hmc' '--pid=Microsoft-MIEngine-P
id-gqx0oh2s.v4v' '--dbgExe=C:\msys64\mingw64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Sorted array: 1 2 4 5 8
PS C:\Users\USER\Documents\Condesa DSA> ^C
PS C:\Users\USER\Documents\Condesa DSA>
PS C:\Users\USER\Documents\Condesa DSA>  ${env:DEBUGINFOD_URLS}=''; & 'c:\Users\USER\.vs
code\extensions\ms-vscode.cpptools-1.34.4-win32-x64\debugAdapters\bin\WindowsDebugLaunch
er.exe' '--stdin=Microsoft-MIEngine-In-age52lir.ows' '--stdout=Microsoft-MIEngine-Out-ey
j4iqpa.ldg' '--stderr=Microsoft-MIEngine-Error-y4jbjdgp.wxc' '--pid=Microsoft-MIEngine-P
id-xpxtmryu.jhf' '--dbgExe=C:\msys64\mingw64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Sorted array: 11 12 22 25 64
PS C:\Users\USER\Documents\Condesa DSA>
```

**Insertion Sort:** Builds the sorted list one element at a time by inserting each new element into its correct position.



**Bubble Sort:** Repeatedly compares adjacent elements and swaps them if they are in the wrong order. Larger elements "bubble up" to the end of the list.



```
Seatwork 7.1 > main.cpp > ...
#include <iostream>
using namespace std;

void bubbleSort(int arr[], int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                swap(arr[j], arr[j + 1]);
            }
        }
    }
}

int main() {
    int arr[] = {5, 1, 4, 2, 8};
    int n = sizeof(arr)/sizeof(arr[0]);
    bubbleSort(arr, n);
    cout << "Sorted array: ";
    for (int i = 0; i < n; i++) cout << arr[i] << " ";
    return 0;
}
```

```
Problems  Output  Debug Console  Terminal  Ports

er.exe' '--stdin=Microsoft-MIEngine-In-el10j3d5.jwi' '--stdout=Microsoft-MIEngine-Out-pk
c4zugx.e0m' '--stderr=Microsoft-MIEngine-Error-f1gsm2o5.rhi' '--pid=Microsoft-MIEngine-P
id-nuwuhn0u.uxj' '--dbgExe=C:\msys64\mingw64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Sorted array: 5 6 11 12 13
PS C:\Users\USER\Documents\Condesa DSA> ^C
PS C:\Users\USER\Documents\Condesa DSA>
PS C:\Users\USER\Documents\Condesa DSA>  ${env:DEBUGINFOD_URLS}=''; & 'c:\Users\USER\.vs
code\extensions\ms-vscode.cpptools-1.34.4-win32-x64\debugAdapters\bin\WindowsDebugLaunch
er.exe' '--stdin=Microsoft-MIEngine-In-5dqk0guf.mz5' '--stdout=Microsoft-MIEngine-Out-dv
w3dlkp.bl4' '--stderr=Microsoft-MIEngine-Error-stmmcply.mie' '--pid=Microsoft-MIEngine-P
id-dhz1nz4z.5hy' '--dbgExe=C:\msys64\mingw64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Sorted array: 1 2 4 5 8
PS C:\Users\USER\Documents\Condesa DSA>
```

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:**

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

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**Purpose of Use:**

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:**

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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